

KANIN NOS, Russian Federation

WMO: 221650

Lat: 68.655N

Lon: 43.288E

Elev: 159

StdP: 14.61

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.9	-3.7	-13.5	2.7	-9.5	-7.3	3.7	-3.2	46.7	15.4	43.5	14.2	15.8	90	1.024

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	64.7	57.4	60.8	55.7	57.4	53.7	58.7	63.1	56.2	60.1	53.9	56.9	14.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.3	68.0	61.4	54.1	62.5	58.3	52.1	58.1	55.8	25.7	63.2	24.0	60.3	22.6	57.0	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.1	35.3	31.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-10.0	61.0	6.5	4.8	-14.7	64.5	-18.5	67.3	-22.1	70.0	-26.9	73.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	32.0	17.5	15.8	19.2	26.0	32.4	41.8	48.1	48.0	44.6	36.3	29.0	24.0	(6)
(7)		DBStd	13.39	9.98	11.17	8.36	5.95	6.14	6.74	6.79	4.63	4.00	4.85	6.49	8.33	(7)
(8)		HDD50	6694	1008	956	954	720	545	267	120	97	169	423	630	805	(8)
(9)		HDD65	12048	1473	1376	1419	1170	1010	697	524	528	613	888	1080	1270	(9)
(10)		CDD50	124	0	0	0	0	1	20	63	34	6	0	0	0	(10)
(11)		CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)
(12)		CDH74	3	0	0	0	0	0	0	3	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	16.3	18.3	19.3	16.9	15.5	14.9	14.2	14.1	13.8	15.2	17.1	17.8	19.3	(14)
(15)	Precipitation	PrecAvg	16.8	1.4	1.1	1.0	0.9	0.9	1.3	1.5	2.0	1.8	2.0	1.6	1.5	(15)
(16)		PrecMax	26.0	4.6	3.3	3.4	2.2	3.4	4.4	3.9	4.7	4.9	4.0	2.9	3.4	(16)
(17)		PrecMin	10.1	0.2	0.0	0.2	0.3	0.2	0.2	0.2	0.1	0.5	0.6	0.5	0.4	(17)
(18)		PrecStd	3.0	0.8	0.6	0.6	0.4	0.5	0.8	0.9	1.1	0.8	0.7	0.6	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.9	33.2	33.5	39.8	54.6	67.3	72.8	64.5	57.4	47.6	42.1	37.2	(19)
(20)			MCWB	32.7	32.6	31.9	35.8	48.1	58.0	62.4	58.4	51.9	45.9	41.2	36.0	(20)
(21)		2%	DB	32.1	31.2	32.2	36.0	48.4	61.7	66.5	60.6	53.3	46.2	40.0	35.4	(21)
(22)			MCWB	31.0	30.5	31.1	33.8	43.8	54.8	58.4	56.7	51.2	44.9	38.9	34.2	(22)
(23)		5%	DB	30.9	29.8	31.1	33.7	43.9	57.4	62.8	57.3	51.4	44.8	38.2	33.9	(23)
(24)			MCWB	30.0	29.0	30.3	32.5	40.9	51.9	57.1	54.8	49.8	43.7	37.0	32.8	(24)
(25)		10%	DB	29.6	28.3	29.9	32.6	40.4	52.8	59.2	54.6	50.0	43.2	36.7	32.7	(25)
(26)			MCWB	28.7	27.4	29.2	31.6	38.3	48.5	55.1	52.5	48.8	41.9	35.6	31.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.0	32.7	32.4	36.2	48.5	60.0	63.4	59.6	53.3	46.4	41.6	36.4	(27)
(28)			MCD B	33.4	32.8	32.8	38.8	53.9	65.0	70.0	63.2	55.0	47.0	41.9	36.8	(28)
(29)		2%	WB	31.3	30.7	31.3	34.0	44.1	55.3	60.0	56.9	51.4	45.1	38.9	34.4	(29)
(30)			MCD B	31.9	31.0	31.9	35.4	47.3	60.5	65.3	60.0	52.9	45.9	39.6	35.1	(30)
(31)		5%	WB	30.1	29.3	30.4	32.9	41.0	51.9	57.6	54.9	50.2	43.8	37.3	33.0	(31)
(32)			MCD B	30.8	29.8	31.1	33.7	43.5	56.9	62.0	57.1	51.4	44.7	38.0	33.7	(32)
(33)		10%	WB	28.8	27.5	29.3	32.0	38.4	48.4	55.2	53.0	48.9	42.2	35.8	31.8	(33)
(34)			MCD B	29.5	28.2	30.0	32.5	40.3	52.9	59.0	54.6	50.0	43.3	36.6	32.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	7.4	7.1	5.9	6.0	7.4	7.4	6.2	4.6	4.3	5.6	6.3	(35)
(36)			MCDBR	5.5	5.7	5.7	6.1	12.0	15.6	12.4	10.9	6.4	4.6	5.0	5.0	(36)
(37)			MCWBR	5.6	5.8	5.1	5.4	9.1	10.9	8.1	8.0	5.2	4.9	5.1	5.2	(37)
(38)		5% WB	MCD BR	5.7	5.6	5.0	5.8	11.5	15.2	11.6	10.1	6.3	4.4	4.8	5.2	(38)
(39)			MCWBR	5.8	5.9	4.9	5.3	9.2	11.3	8.4	7.8	5.3	4.9	5.1	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.375	0.269	0.307	0.334	0.339	0.336	0.366	0.345	0.321	0.289	0.361	N/A	(40)	
(41)		taud	2.570	2.066	2.151	2.250	2.416	2.518	2.457	2.502	2.504	2.220	2.540	N/A	(41)	
(42)		Ebn at Noon	26	153	223	255	270	275	260	253	229	147	21	0	(42)	
(43)		Edh at Noon	4	15	25	31	30	28	29	25	19	13	3	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6	103	504	1002	1373	1630	1476	972	531	150	17	0	(44)	
(45)		RadStd	1	22	57	87	105	194	179	105	63	23	2	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page